INCH-POUND

MIL-DTL-0085097/13A(OS)
28 August 1996
USED IN LIEU OF
MIL-A-85097/13(AS)
24 August 1984

MILITARY SPECIFICATION SHEET

AIRCREW ESCAPE PROPULSION SYSTEMS (AEPS)
ROCKET MOTOR, CANOPY JETTISON, MARK 109 MOD 0 (LOADED ASSEMBLY)

This specification sheet is approved for use within the Naval Sea Systems Command, Department of the Navy, based upon currently available technical information but it has not been approved for promulgation as a coordinated revision of MIL-A-85097/13. It is subject to modification. However, pending its promulgation as a coordinated military specification, it may be used in acquisition.

The requirements for acquiring the Mark 109 Mod 0 Canopy Jettison Rocket Motor Loaded Assembly described herein shall consist of this specification sheet and the issue of MIL-A-85097 listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation.

1. SCOPE

1.1 Scope. This specification covers the requirements unique to the Canopy Jettison Rocket Motor Mark 109 Mod 0 Loaded Assembly (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standard, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Indian Head Division, Naval Surface Warfare Center, Standardization Branch (Code 8420), 101 Strauss Avenue, Indian Head, MD 20640-5035, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by sending a letter.

AMSC N/A FSC 1377

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SPECIFICATIONS

MILITARY

MIL-B-5087	Bonding, Electrical, and Lightning Protection, For Aerospace Systems
MIL-P-83126	Propulsion Systems, Aircrew Escape, Design Specification For

2.2.2 Other Government documents, drawings, and publications. The following drawing forms a part of this document to the extent specified herein. Unless otherwise specified, the issue is that cited in the solicitation (see 6.2).

DRAWINGS

NAVAL AIR SYSTEMS COMMAND (CAGE Code 30003)

1507AS100	Rocket Motor, Canopy Jettison, Mk 109 Mod 0

(Application for copies of specifications and drawings should be addressed to the Commander, Indian Head Division, Naval Surface Warfare Center, Attn: Code 8410P, 101 Strauss Avenue, Indian Head, MD 20640-5035.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 General. The requirements in this specification sheet, unless in direct conflict with the basic specification, shall be considered additional, not substitute. The Rocket Motor described herein shall conform to Drawing 1507AS100 and the weight shall be not greater than 3.50 pounds.

3.2 Preproduction.

3.2.1 Propellant aging. When tested in accordance with MIL-A-85097, the propellant shall meet the aging limit requirements specified by the contractor and which are acceptable to the Government.

3.3 Preproduction/production.

3.3.1 Propellant. The propellant shall conform to the contractor's in-house propellant specification when tested for the data of MIL-A-85097. The contractor shall make available the results of the tested values and the in-house specification limits for comparison (see 6.3).

3.3.2 Ballistic performance requirements. The ballistic performance requirements shall conform to table I at both $-65 \pm 5^{\circ}$F and $165 \pm 5^{\circ}$F.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Preproduction/production.

4.1.1 Conditioning temperature. The conditioning temperatures and tolerances shall be as follows:

a. Cold: $-65 \pm 5^{\circ}F$
b. Hot: $165 \pm 5^{\circ}F$

TABLE I. Ballistic performance requirements.

Characteristic	Minimum	Maximum
Total impulse (lb-s)	184	212
Action time (ms)	-	170
Thrust (lb)	-	4300
Time to 70% maximum rocket motor thrust (ms)	-	20

4.1.2 Conditioning period. Time for temperature conditioning prior to static firing or during temperature cycling shall be as follows:

a. Static firing: 4 hours, minimum
b. Temperature cycling: 5 hours, minimum

4.1.3 Radiographic inspection. All components or assemblies shall be radiographed in two views 90° apart.

4.1.4 Hydrostatic proof test. Each motor case shall be hydrostatically tested as a pressure vessel to 6366 psi for 15 seconds.

4.1.5 Leakage test. Each rocket motor assembly shall be subjected to a dry leak test to verify compliance with a leakage rate of less than 1×10^{-4} cc/sec air.

4.1.6 Threaded joint torque test. The 1.890-24NS-3 threaded joints (2) of each rocket motor assembly shall withstand 400-inch-pounds torque without breakaway, when tested in accordance with MIL-P-83126.

4.1.7 Electrical bonding resistance test. For each rocket motor assembly selected as a sampling test specimen, the electrical resistance between the shielded mild detonating cord (SMDC) housing and the mounting lugs shall be not greater than 0.0025 ohms, when tested in accordance with MIL-B-5087.

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5. PACKAGING

(This section is not applicable to this specification sheet.)

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The rocket motor described herein is intended to be used to eject the canopy during an emergency crew ejection from the F/A/TF-18 Aircraft.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.

b. Issue of DoDISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.2.1 and 2.2.2).

c. Mark and Mod designation of rocket motor.

6.3 Consideration of data requirements. The following data requirements should be considered when this specification is applied on a contract. The applicable Data Item Descriptions (DID's) should be reviewed in conjunction with the specific acquisition to ensure that only essential data are requested/provided and that the DID's are tailored to reflect the requirements of the specific acquisition. To ensure correct contractual application of data requirements, a Contract Data Requirements List (DD Form 1423) must be prepared to obtain the data, except where DOD FAR Supplement 27.475-1 exempts the requirement for a DD Form 1423.

Reference paragraph	DID number	DID title	Suggested tailoring
3.3.1	UDI-T-22741A	Acceptance Test Report	—

The above DID's were those cleared as of the date of this specification. The current issue of DOD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL), must be researched to ensure that only current, cleared DID's are cited on the DD Form 1423.

6.4 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Preparing activity:
Navy - OS
(Project 1377-0009)

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